



KWAZULU-NATAL PROVINCE

EDUCATION
REPUBLIC OF SOUTH AFRICA

**NATIONAL
SENIOR CERTIFICATE**

GRADE 11

MATHEMATICS P1

NOVEMBER 2025

Stanmorephysics.com

Stanmorephysics.com

MARKS: 150

TIME: 3 hours

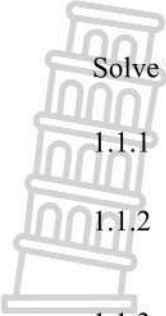
This question paper consists of 9 pages and 1 information sheet.

INSTRUCTIONS AND INFORMATION

Read the following instructions carefully before answering the questions.

1. This question paper consists of 10 questions.
2. Answer ALL the questions in the SPECIAL ANSWER BOOK provided.
3. Clearly show numbering if additional space has been used to answer some question. If NOT, cross out all rough work done in the additional space.
4. Clearly show ALL calculations, diagrams, graphs, etc. which you have used in determining your answers.
5. Answers only will NOT necessarily be awarded full marks.
6. You may use an approved scientific calculator (non-programmable and non-graphical), unless stated otherwise.
7. If necessary, round off answers correct to TWO decimal places, unless stated otherwise.
8. Diagrams are NOT necessarily drawn to scale.
9. An information sheet with formulae is included at the end of the question paper.
10. Write neatly and legibly.

QUESTION 11.1 Solve for x :


 1.1.1 $4x^2 - 9 = 0$ (3)

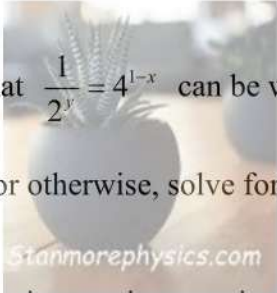
1.1.2 $-2x^2 + 6x - 3 = 0$ (answers correct to TWO decimal places) (4)

1.1.3 $x - 3 = \frac{2x^2 + x - 6}{x + 2}$ (4)

1.1.4 $x^2 - 2x - 15 \geq 9$ (4)

1.2 Two equations are given below:

$$\frac{1}{2^y} = 4^{1-x} \quad \text{and} \quad x + 6 = x^2 - y$$


 1.2.1 Show that $\frac{1}{2^y} = 4^{1-x}$ can be written as $y = 2x - 2$. (2)

1.2.2 Hence, or otherwise, solve for both x and y . (4)

 1.3 The roots of a quadratic equation are given as: $x = \sqrt{2-k}$, where $k \in (-9 : 0)$. Determine the integral values of k for which the roots are rational. (2)

 1.4 The roots of the equation $ax^2 + bx + c = 0$ are $\frac{-b \pm \sqrt{72}}{6}$ where $b < 0$ and $c = b + 18$. Calculate the values of a , b and c . (5)
[27]

QUESTION 2

2.1 Simplify fully, **WITHOUT** using a calculator:



2.1.1 $2x \cdot (27x^3)^{\frac{2}{3}} \cdot \left(\frac{2x}{27x^3}\right)^0$ (3)

2.1.2 $\frac{(6-2\sqrt{3})^2}{\sqrt{3}}$ (4)

2.2 It is given that $\frac{2^{m+4} - 4 \cdot 2^{m+1}}{5 \cdot 2^{m+3}}$

2.2.1 Simplify the above expression. (3)

2.2.2 Hence, solve for x if $\frac{5 \cdot 2^{m+3}}{2^{m+4} - 4 \cdot 2^{m+1}} = \frac{1}{125^x}$ (3)

2.3 Solve for x : $\sqrt{x^2 - 53} - 2\sqrt{2x - 5} = 0$ (5)
[18]

QUESTION 3

WhatsApp messages were sent using face emojis with two popping hearts per face (shown below).

MESSAGE 1	MESSAGE 2	MESSAGE 3

These faces and popping hearts in the different messages forms a linear number pattern.

3.1 How many popping hearts will appear in message 4? (1)

3.2 Determine the n^{th} term (general term) of the popping hearts in the messages. (2)

3.3 Calculate the number of popping hearts in the tenth (T_{10}) message. (2)

3.4 In which message will there be 710 popping hearts? (2)

3.5 A new series of messages using the same emojis was created by:

- keeping the number of faces in message 1 and message 2 the same as the original messages.
- adding 2 more faces to the original message 3.
- adding 6 more faces to the original message 4.

The new series of messages forms a quadratic number pattern.

Which message in the NEW PATTERN will have 266 popping hearts ?

(6)
[13]

QUESTION 4

The first four terms of a quadratic number pattern are -7 ; m ; 5 ; n . The constant second difference is 4 .

4.1 Prove, by calculations, that $m = -3$ and $n = 17$. (3)

4.2 Which two consecutive terms of the quadratic number pattern has a first difference of 292 ? (3)

4.3 It is given that the general term of a quadratic number pattern is $T_n = 2n^2 - 2n - 7$. Calculate the value of n if $T_n = 1\,005$. (4)
[10]

QUESTION 5

Consider the function f defined by $f(x) = \frac{-3}{x+2} + \frac{1}{2}$. Graph f passes through $(4 ; 0)$.

5.1 Write down the y -intercept of f . (1)

5.2 Write down the equations of the asymptotes of f . (2)

5.3 Calculate the x -intercept of f . (2)

5.4 Draw a neat sketch of f . Clearly label all the intercepts with the axes and asymptotes. (3)

5.5 Determine the values of y for which f lies $x < -2$ or $0 \leq x \leq 4$. (2)

5.6 Graph f is reflected about the y -axis and shifted $\frac{3}{2}$ units up to form p .
Write down the equation of p in the form of $p(x) = \dots$. (2)
[12]

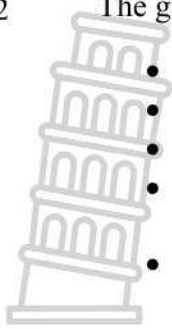
QUESTION 6

6.1 The function $h(x) = ax^2 + bx + c$ has the following properties:

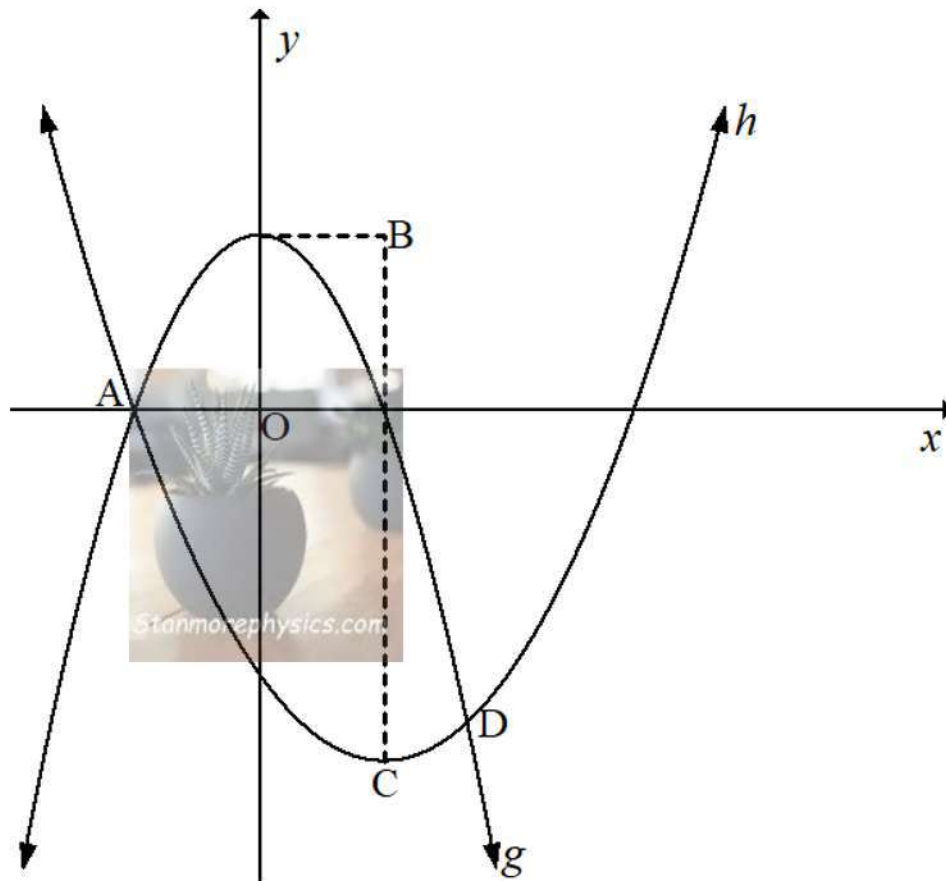
- $h(2) = 0$
- the value of $\frac{-b}{2a}$ is 5
- the range of h is $y \leq 10 ; y \in \mathbb{R}$.

Draw a neat sketch of h . Clearly label the x -intercepts and turning point. (3)

6.2 The graphs of $g(x) = -x^2 + 9$ and $h(x) = ax^2 + bx - \frac{27}{4}$ are sketched below.



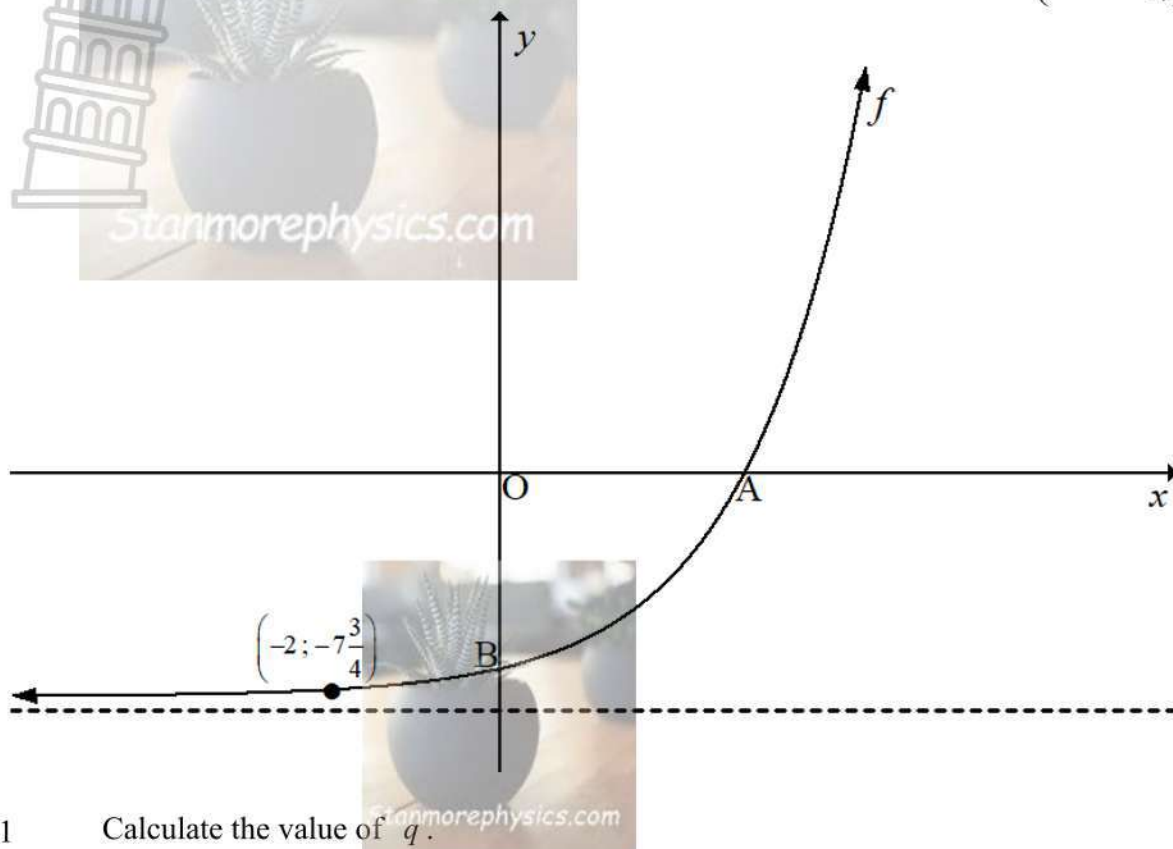
- A is an x -intercept of g and h and lies on the negative x -axis.
- A and D are the points of intersection of graphs g and h .
- C is the turning point of h .
- B lies on the horizontal line that passes through the turning point of g such that BC is parallel to the y -axis and passes through the other intercept of g .
- $BC = 18$ units



- 6.2.1 Calculate the coordinates of A. (3)
- 6.2.2 Show that $a = \frac{1}{4}$ and $b = -\frac{3}{2}$. (4)
- 6.2.3 Write down the range of h . (2)
- 6.2.4 It is given that $m(x) = x^2 - 4$. Explain, in words, the transformation from the graphs of g to m . (2)
- 6.2.5 Calculate the coordinates of D. (5)
- 6.2.6 It is given that $p(x) = k$. Determine the integral value of k such that the distance between the points of intersection of h and p is 8 units. (2)

QUESTION 7

The graph of $f(x) = 2^x + q$ is drawn below. . Graph f passes through the point $\left(-2; -7\frac{3}{4}\right)$.



- 7.1 Calculate the value of q . (2)
- 7.2 If $q = -8$, calculate the coordinates of A. (3)
- 7.3 Determine the average gradient between A and B. (3)
- 7.4 Determine the values of x where $\frac{x}{f(x)} \leq 0$ (2)
- 7.5 Determine the values of k for which $8 - 2^x - k = 0$ will have negative roots. (2)

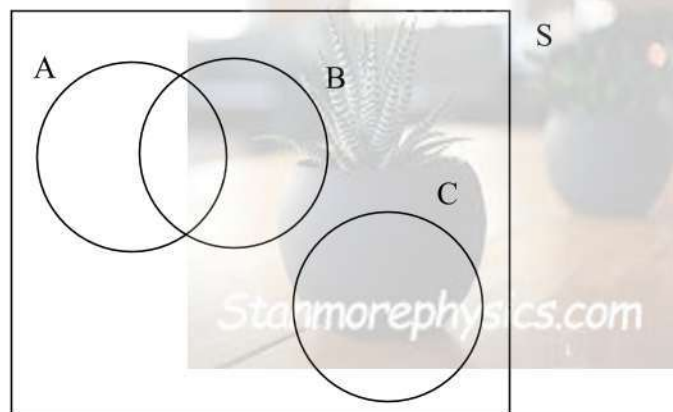
[12]

QUESTION 8

- 8.1 Dr Krishnanair invests R15 000 for a period of 6 years in an account that pays simple interest at the rate of 10% per annum. Calculate the interest his investment earned at the end of 6 years. (3)
- 8.2 Miss Coetzee bought a brand new car. The value of the car depreciates annually at the rate of 8,25% p.a. on the reducing balance method. If the book value of the car was R277 216 after 3 years, calculate the purchase price of the car. (2)
- 8.3 An advertisement reads "XY Bank offers loans at an effective interest rate of 12% per annum". Calculate the nominal rate of interest if interest on the loan is compounded quarterly. (4)
- 8.4 Mr Dlezi invested R25 000 in an account that offers 9,25% per annum compounded half-yearly. Calculate the accumulated amount in this account at the end of nineteen half-years. (3)
- 8.5 Mr Masango invested R6 500 in a savings account that paid interest at the rate of 13% p.a., compounded monthly for the first two years. Thereafter, the interest rate changed to 10,5% p.a., compounded quarterly. Three years after making the initial deposit, Mr Masango withdrew Rx from the account. At the end of 5 years, he had R928,30 more in the account than his initial deposit. Calculate the value of x , to the nearest rand (5)
- [17]

QUESTION 9

In the Venn diagram below, events A, B and C are shown.



9.1 It is given that $P(A) = 0,47$ and $P(C) = \frac{3}{7}$

9.1.1 Write down the $P(A \text{ and } C)$. (1)

9.1.2 Calculate the $P(A \text{ or } C)$. (2)

9.2 It is given that events A and B are independent and $P(A) = \frac{2}{5}P(B)$.

9.2.1 Determine the $P(A \text{ and } B)$ in terms of $P(B)$. (2)

9.2.2 It also given that $P(A \text{ or } B) = 70\%$. Calculate $P(B)$. (5)

[10]

QUESTION 10

A survey was conducted amongst 45 males and 85 females who were randomly selected. This survey was about the brand of clothes they would love to buy between Gucci (G) and Louis Vuitton (LV). The results are summarised below.

- 3y females said they would love to buy Louis Vuitton.
- y males said they would love to buy Gucci.
- The total number of males who said they would love to buy Louis Vuitton is double the total number of females who said they would love to buy Gucci.

10.1 Calculate the total number of females who said they would love to buy Gucci. (6)

10.2 Are the events of being a 'male' and having said that they would love to buy one of the brands **independent**? Justify your answer with calculations. (4)

[10]

TOTAL MARKS: 150

INFORMATION SHEET

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$A = P(1 + ni)$$

$$A = P(1 - ni)$$

$$A = P(1 - i)^n$$

$$A = P(1 + i)^n$$

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(r^n - 1)}{r - 1}; r \neq 1$$

$$S_\infty = \frac{a}{1 - r}; -1 < r < 1$$

$$F = \frac{x[(1 + i)^n - 1]}{i}$$

$$P = \frac{x[1 - (1 + i)^{-n}]}{i}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$M\left(\frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2}\right)$$

$$y = mx + c$$

$$y - y_1 = m(x - x_1)$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \tan \theta$$

$$(x - a)^2 + (y - b)^2 = r^2$$

$$\text{In } \triangle ABC: \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A$$

$$\text{area } \triangle ABC = \frac{1}{2} ab \cdot \sin C$$

$$\sin(\alpha + \beta) = \sin \alpha \cdot \cos \beta + \cos \alpha \cdot \sin \beta$$

$$\sin(\alpha - \beta) = \sin \alpha \cdot \cos \beta - \cos \alpha \cdot \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cdot \cos \beta - \sin \alpha \cdot \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cdot \cos \beta + \sin \alpha \cdot \sin \beta$$

$$\cos 2\alpha = \begin{cases} \cos^2 \alpha - \sin^2 \alpha \\ 1 - 2\sin^2 \alpha \\ 2\cos^2 \alpha - 1 \end{cases}$$

$$\sin 2\alpha = 2 \sin \alpha \cdot \cos \alpha$$

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

$$\hat{y} = a + bx$$

$$b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$



KWAZULU-NATAL PROVINCE

EDUCATION
REPUBLIC OF SOUTH AFRICA

NATIONAL SENIOR CERTIFICATE

GRADE 11

MATHEMATICS P1

NOVEMBER 2025

ANSWER BOOK

Stanmorephysics.com

Stanmorephysics.com

Full Name & Surname

For Educator / Moderator Use Only (NOT for learners to complete)

QUESTION	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
Total Per Question	27	18	13	10	12	21	12	17	10	10
Mark Scored										
Moderated Mark										

GRAND TOTAL

150

This answer book consists of 18 pages.

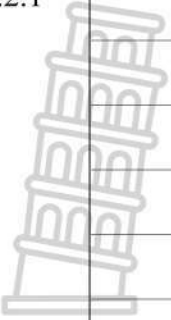
QUESTION 1

	Solution	Marks
1.1.1		(3)
1.1.2		(3)
1.1.3		(4)
1.1.4		(4)

1.2.1		(2)
1.2.2		(4)
1.3		(2)
1.4		(5)
		[27]

QUESTION 2

	Solution	Marks
2.1.1		
2.1.2		(3)
		(4)

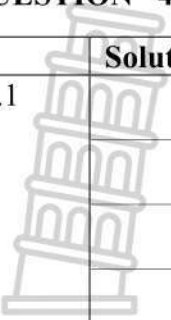
2.2.1		(3)
2.2.2		(3)
2.3		(5)
		[18]

QUESTION	Solution
1.	
2.	

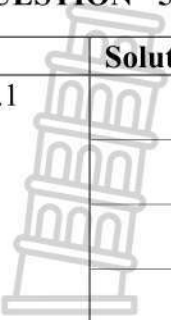
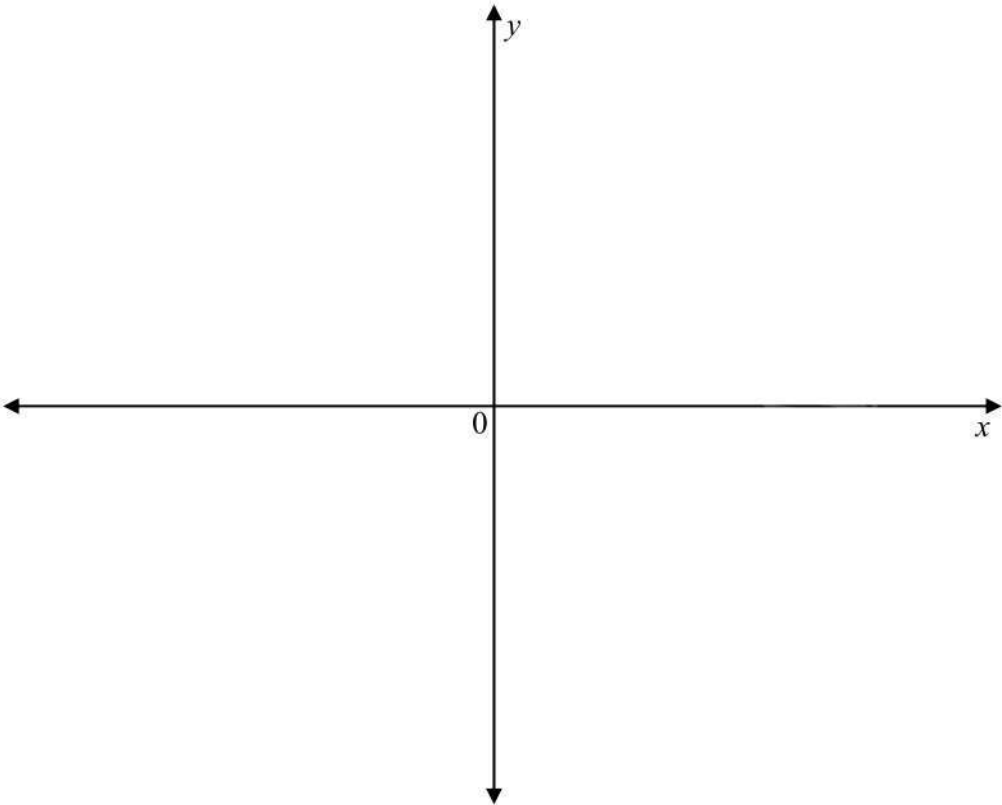


stanmorephysics.com

QUESTION 4

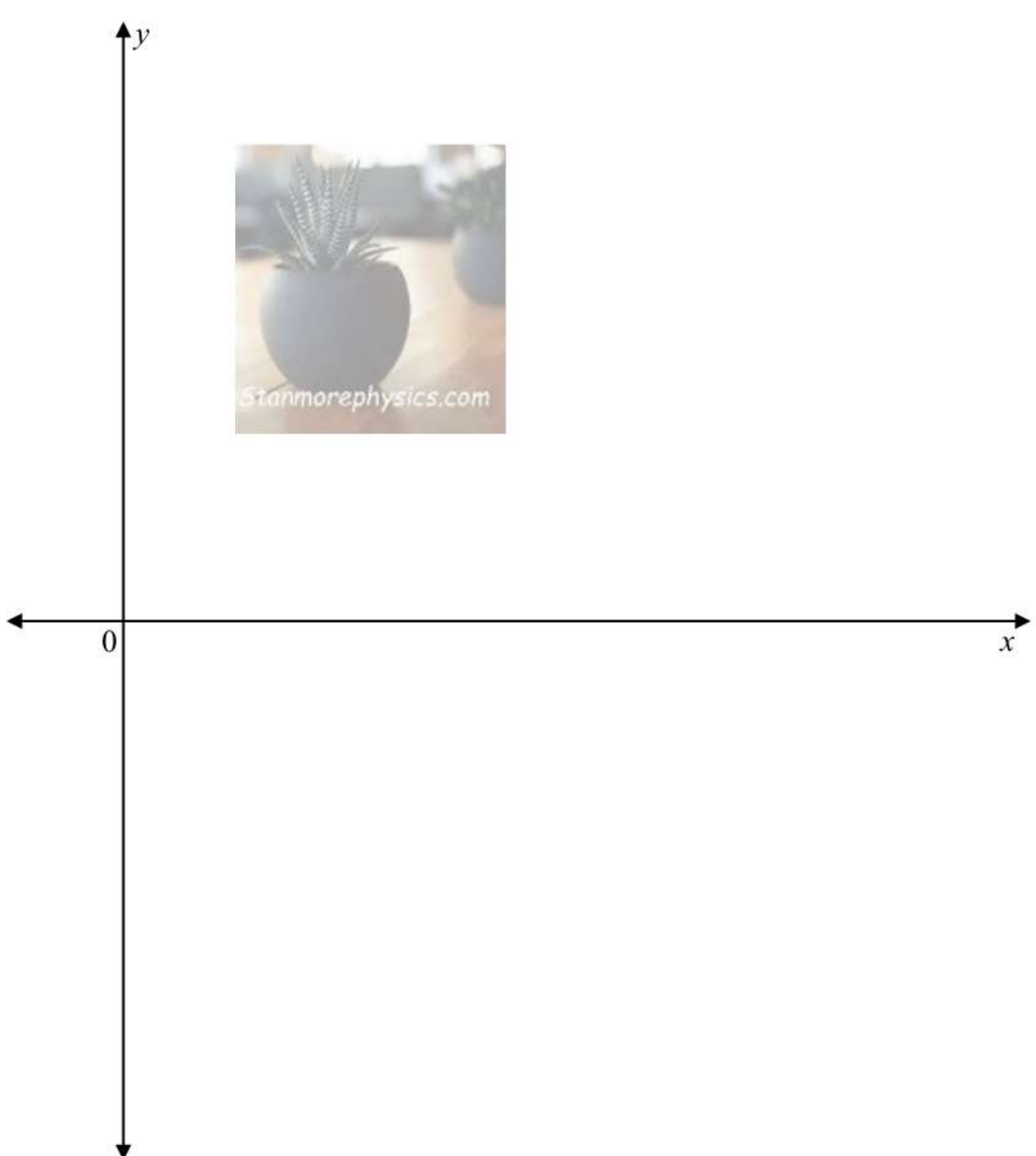
	Solution	Marks
4.1		
		(3)
4.2		
		(3)
4.3		
		(4)
		[10]

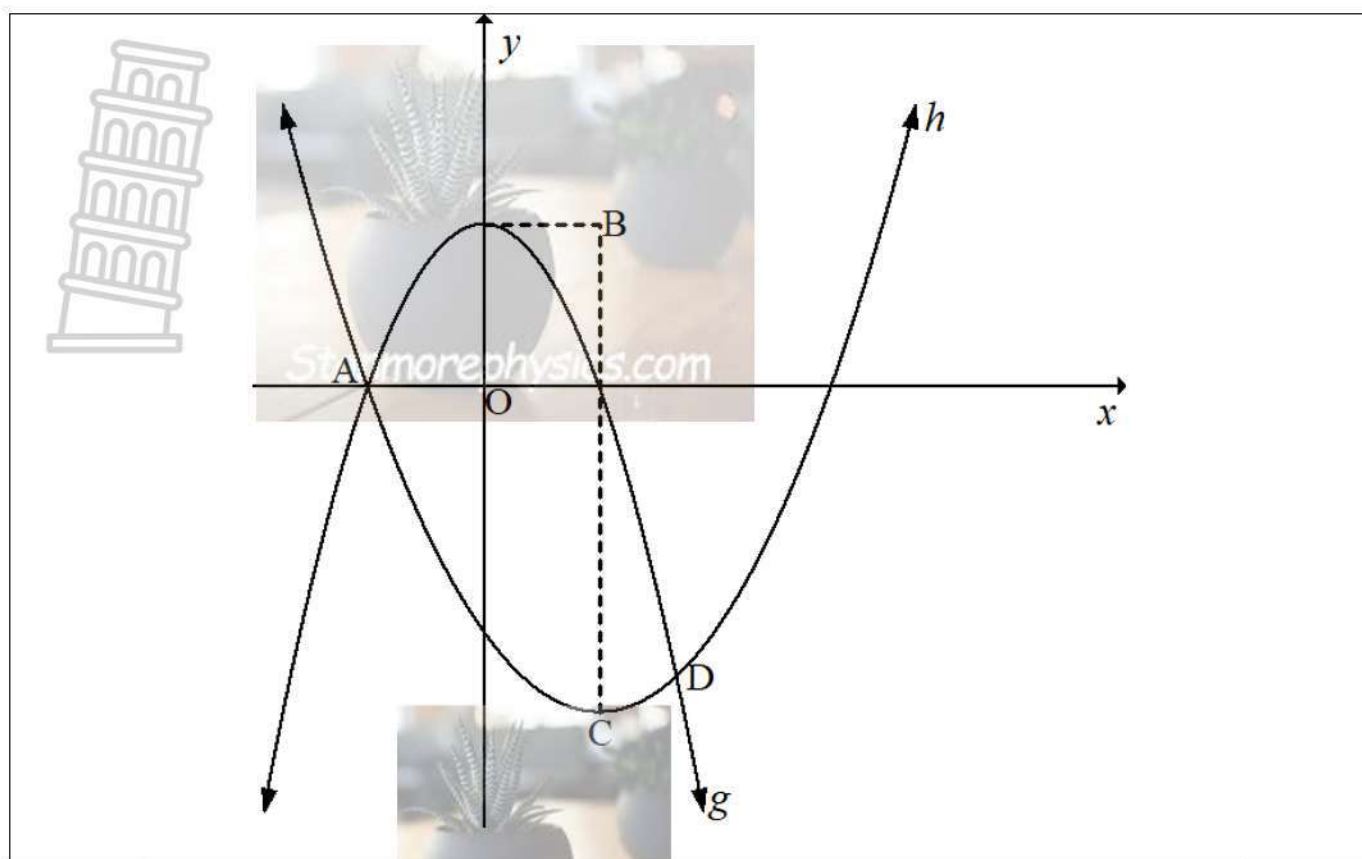
QUESTION 5

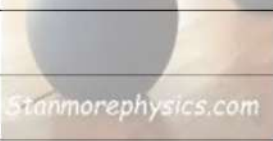
	Solution	Marks
5.1		(1)
5.2		(2)
5.3		(2)
5.4		(3)

5.5		(2)
5.6		(2)
		[12]

QUESTION 6

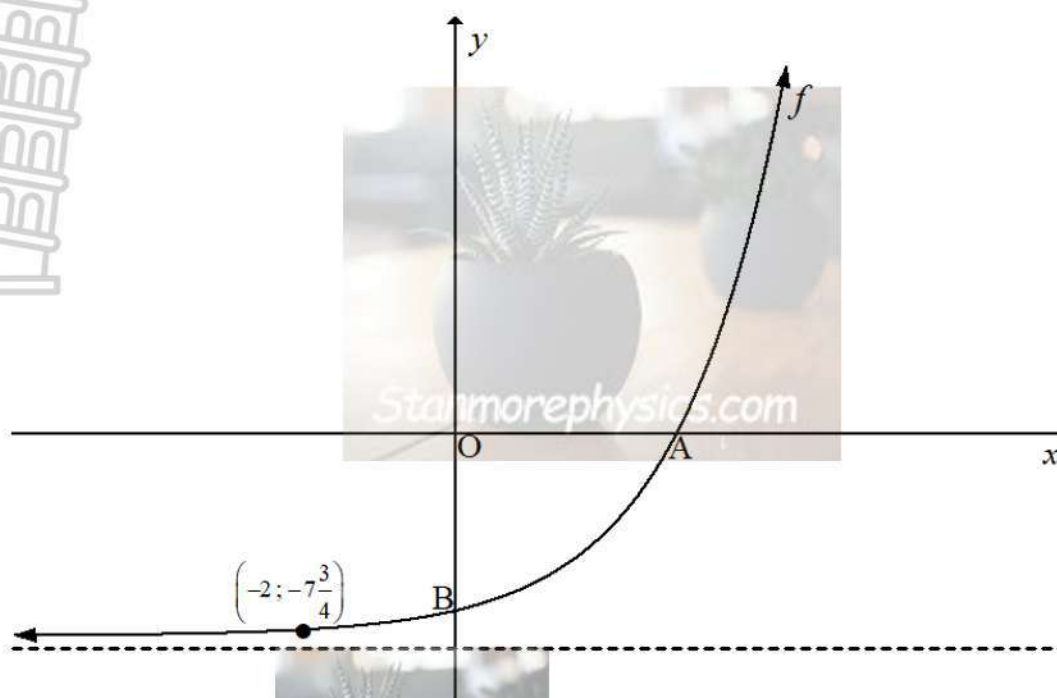
	Solution	Marks
6.1		(3)



6.2.1		(3)
6.2.2		(4)

6.2.3		(2)
6.2.4		(2)
6.2.5		(5)
6.2.6		(2)
		[21]

QUESTION 7



	Solution	Marks
7.1		(2)
7.2		(3)
7.3		(3)

7.4		
		(2)
7.5		
		(2)
		[12]

QUESTION 8

	Solution	Marks
8.1		
		(3)
8.2		
		(2)
8.3		
		(4)

8.4		(3)
8.5		(5)
		[17]

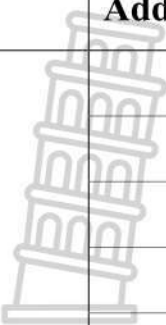
QUESTION 9

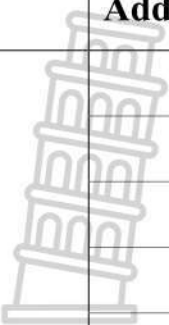
	Solution	Marks
9.1.1		
		(1)
9.1.2		
		(2)
9.2.1		
		(2)
9.2.2		
		(5)
		[10]

QUESTION 10

	Solution	Marks
10.1		(6)
10.2		(4)
		[10]

TOTAL : 150

Additional space		Marks
 		

Additional space		Marks
 		



education

Department:
Education
PROVINCE OF KWAZULU-NATAL

**FINAL & CLOSED
MARKING
GUIDELINES**

MATHEMATICS P1

COMMON EXAM

NOVEMBER 2025

MARKING GUIDELINES

Stanmorephysics.com

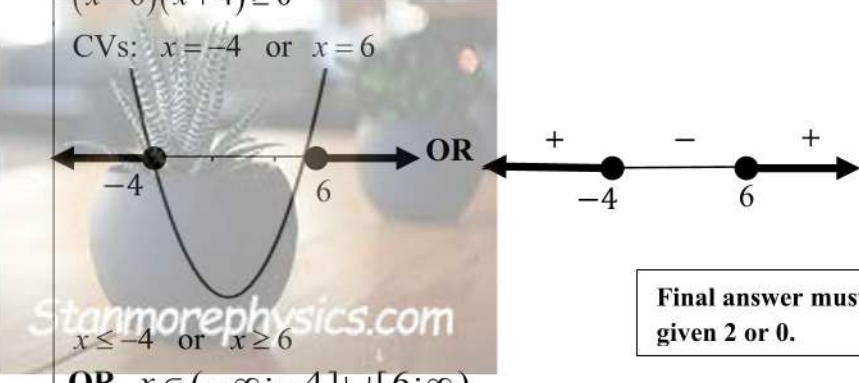
**NATIONAL
SENIOR CERTIFICATE**

GRADE 11

MARKS: 150

These marking guidelines consist of 19 pages.

QUESTION 1

1.1.1	$4x^2 - 9 = 0$ $(2x+3)(2x-3) = 0$ $x = -\frac{3}{2}$ or $x = \frac{3}{2}$ Accept subst. to quadratic formula OR $4x^2 - 9 = 0$ $\sqrt{x^2} = \pm \sqrt{\frac{9}{4}}$ $x = -\frac{3}{2}$ or $x = \frac{3}{2}$	✓ A correct factors ✓ CA answer ✓ CA answer (3) OR ✓ A isolating x^2 ✓ CA answer ✓ CA answer (3)
1.1.2	$-2x^2 + 6x - 3 = 0$ $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(-2)(-3)}}{2(-2)}$ OR $x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(2)(3)}}{2(2)}$ $x = 2,37$ or $x = 0,63$ Penalise 1 mark for incorrect rounding	✓ A correct subst. into quadratic formula ✓ CA answer ✓ CA answer (3)
1.1.3	$x - 3 = \frac{2x^2 + x - 6}{x + 2}$ $(x - 3)(x + 2) = 2x^2 + x - 6$ $x^2 - x - 6 = 2x^2 + x - 6$ $2x^2 + 2x = 0$ $x(x + 2) = 0$ $x = 0$ or $x \neq -2$	✓ A multiplication ✓ CA standard form ✓ CA answer ✓ A rejection (4)
1.1.4	$(x - 5)(x + 3) \geq 9$ $x^2 - 2x - 15 - 9 \geq 0$ $x^2 - 2x - 24 \geq 0$ $(x - 6)(x + 4) \geq 0$ CVs: $x = -4$ or $x = 6$  OR $\begin{matrix} + & - & + \\ \bullet & \bullet & \bullet \\ -4 & & 6 \end{matrix}$ $x \leq -4$ or $x \geq 6$ OR $x \in (-\infty; -4] \cup [6; \infty)$ Final answer must be given 2 or 0.	✓ A standard form ✓ CA critical value (accept if shown on a sketch) ✓✓ CA both answers (4)

CAPS – Grade 11
Marking Guidelines

1.2.1	$\frac{1}{2^y} = 4^{1-x}$ $2^{-y} = 2^{2(1-x)}$ $2^{-y} = 2^{2-2x}$ $-y = 2 - 2x$ $\therefore y = 2x - 2$ 1.2.1 is a proof, all marks are accuracy.	✓A $2^{-y} = 2^{2(1-x)}$ ✓A $-y = 2 - 2x$ (2)
1.2.2	$x+6 = x^2 - y$ $x+6 = x^2 - (2x-2)$ $x^2 - 3x - 4 = 0$ $(x-4)(x+1) = 0$ $x = 4 \text{ or } x = -1$ $y = 2(4) - 2 \text{ or } y = 2(-1) - 2$ $y = 6 \text{ or } y = -4$  OR $x+6 = x^2 - y$ $y = x^2 - x - 6$ $(2x-2) = x^2 - x - 6$ $x^2 - 3x - 4 = 0$ $(x-4)(x+1) = 0$ $x = 4 \text{ or } x = -1$ $y = 2(4) - 2 \text{ or } y = 2(-1) - 2$ $y = 6 \text{ or } y = -4$	✓A substitution ✓CA factors ✓CA both values of x ✓CA both values of y OR ✓A substitution ✓CA factors ✓CA both values of x ✓CA both values of y (4)
1.3	$\Delta = 2 - (-2) = 4 \quad \& \quad \Delta = 2 - (-7) = 9$ $k = -2 \quad \text{or} \quad k = -7$	✓A $k = -7$ ✓A $k = -2$ (2)
1.4	$2a = 6$ $a = 3$ $b^2 - 4ac = 72$ $b^2 - 4(3)(b+18) = 72$ $b^2 - 12b - 216 = 72$ $b^2 - 12b - 288 = 0$ $(b-24)(b+12) = 0$ $b \neq 24 \text{ or } b = -12$ $c = -12 + 18$ $c = 6$	✓A value of a ✓CA subst. a & c into $\Delta = 72$ ✓CA standard form ✓CA value of b with rejection ✓CA value of c (5)
[27]		

QUESTION 2

2.1.1	$2x \cdot (27x^3)^{\frac{2}{3}} \cdot \left(\frac{2x}{27x^3}\right)^0$ $= 2x \cdot (3^3 x^3)^{\frac{2}{3}} \times 1$ $= 2x \cdot (3x)^{3 \times \frac{2}{3}}$ $= 2x(3x)^2$ $= 2x(3^2 x^2)$ $= 18x^3$	✓A 1 ✓A $3^2 x^2$ ✓CA answer (3)
2.1.2	$\frac{(6-2\sqrt{3})^2}{\sqrt{3}}$ $= \frac{36 - 24\sqrt{3} + 12}{\sqrt{3}}$ $= \frac{48 - 24\sqrt{3}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$ $= \frac{48\sqrt{3} - 24 \times 3}{3}$ $= \frac{3(16\sqrt{3} - 24)}{3}$ $= 16\sqrt{3} - 24$	✓A removing the brackets in numerator ✓A rationalising denom. i.e. $\times \frac{\sqrt{3}}{\sqrt{3}}$ ✓CA simplifying the numerator ✓CA answer (4)
2.2.1	$\frac{2^{m+4} - 4 \cdot 2^{m+1}}{5 \cdot 2^{m+3}}$ $= \frac{2^m \cdot 2^4 - 4 \cdot 2^m \cdot 2^1}{2^m \cdot 5 \cdot 2^3} \quad \text{OR} \quad = \frac{2^m \cdot 2^4 - 2^2 \cdot 2^m \cdot 2^1}{2^m \cdot 5 \cdot 2^3}$ $= \frac{2^m (2^4 - 4 \times 2)}{2^m \cdot 5 \cdot 8} \quad \text{OR} \quad = \frac{2^m \cdot 2^3 (2^1 - 1)}{2^m \cdot 2^3 \cdot 5}$ $= \frac{8}{40} \text{ or } \frac{1}{5} \text{ or } 0,2$ OR	✓A expansion using exponent law ✓A factorising the numerator ✓CA answer OR ✓✓A factorising the numerator (3)

CAPS – Grade 11
Marking Guidelines

	$\frac{2^{m+4} - 4 \cdot 2^{m+1}}{5 \cdot 2^{m+3}}$ $= \frac{2^m(2^4 - 4 \times 2)}{2^m \cdot 5 \cdot 8} \quad \text{OR} \quad = \frac{2^m \cdot 2^3(2^1 - 1)}{2^m \cdot 2^3 \cdot 5}$ $= \frac{8}{40} \text{ or } \frac{1}{5} \text{ or } 0,2$ OR $\frac{2^{m+4} - 4 \cdot 2^{m+1}}{5 \cdot 2^{m+3}}$ $= \frac{2^m \cdot 16 - 8 \cdot 2^m}{5 \cdot 2^m \cdot 8}$ $= \frac{2^m \cdot 8}{5 \cdot 2^m \cdot 8}$ $= \frac{8}{40} \text{ or } \frac{1}{5} \text{ or } 0,2$ OR $\frac{2^{m+4} - 4 \cdot 2^{m+1}}{5 \cdot 2^{m+3}}$ $= \frac{2^{m+4}}{2^m \cdot 5 \cdot 8} - \frac{4 \cdot 2^{m+1}}{2^m \cdot 5 \cdot 8}$ $= \frac{2^{m+4-m}}{40} - \frac{4 \cdot 2^{m+1-m}}{40}$ $= \frac{16}{40} - \frac{8}{40}$ $= \frac{8}{40} \text{ or } \frac{1}{5} \text{ or } 0,2$	✓CA answer (3) OR ✓A simplifying terms in numerator ✓A adding like terms in numerator ✓CA answer (3) OR ✓A splitting to two fractions ✓A simplification ✓CA answer (3)
2.2.2	$5 = \frac{1}{125^x}$ $5 = \frac{1}{5^{3x}}$ $5 \cdot 5^{3x} = 1$ $5^{1+3x} = 5^0$ $1 + 3x = 0$ $x = -\frac{1}{3}$ OR	✓A equating $\frac{1}{125^x}$ to 5 ✓A $5^{1+3x} = 5^0$ ✓CA answer (3) OR ✓A equating $\frac{1}{125^x}$ to 5

CAPS – Grade 11
Marking Guidelines

	$5 = \frac{1}{125^x}$ $5 = \frac{1}{5^{3x}}$ $5 = 5^{-3x}$ $1 = -3x$ $x = -\frac{1}{3}$	✓A $5 = 5^{-3x}$ ✓CA answer (3)
2.3	$\sqrt{x^2 - 53} - 2\sqrt{2x - 5} = 0$ $\sqrt{x^2 - 53} = 2\sqrt{2x - 5}$ $(\sqrt{x^2 - 53})^2 = (2\sqrt{2x - 5})^2$ $x^2 - 53 = 4(2x - 5)$ $x^2 - 53 = 8x - 20$ $x^2 - 8x - 33 = 0$ $(x - 11)(x + 3) = 0$ $x = 11 \text{ or } x = -3$	✓A isolating surds ✓A squaring on both sides ✓CA standard form ✓CA answers with selection. (4)
[17]		

QUESTION 3

3.1.	14	✓A answer (1)
3.2	2 ; 6 ; 10 ; 14 $T_n = 4n - 2$	✓A $4n$ ✓A -2 (2)
3.3	$T_{10} = 4(10) - 2$ or $T_5 = [2(10) - 1] \times 2$ $T_{10} = 38$ Answer only full marks	✓CA subst. of $n = 10$ ✓CA answer (2)
3.4	$T_n = 4n - 2$ $710 = 4n - 2$ $4n = 712$ $n = 178$ OR Answer only full marks OR If n is a decimal, it must be rounded to whole number	✓CA $T_n = 710$ ✓CA answer OR ✓CA $T_n = 355$ (2)

CAPS – Grade 11
Marking Guidelines

	$2n - 1 = 355$ $2n = 356$ $n = 178$	Answer only full marks OR If n is a decimal, it must be rounded to whole number	✓CA answer (2)
3.5	2 ; 6 ; 14 ; 26.....hearts quad seq 4 ; 8 ; 12.....first difference 4 ; 4second difference $2a = 4$ $3(2) + b = 4$ $2 - 2 + c = 2$ $a = 2$ $b = -2$ $c = 2$ $T_n = 2n^2 - 2n + 2$ $266 = 2n^2 - 2n + 2$ $2n^2 - 2n + 2 - 266 = 0$ $n^2 - n - 132 = 0$ $(n - 12)(n + 11) = 0$ $n = 12$ or $n \neq -11$ Only the 12th message. OR 1 ; 3 ; 7 ; 13.....faces quad seq 2 ; 4 ; 6.....first difference 2 ; 2second difference $2a = 2$ $3(1) + b = 4$ $1 - 1 + c = 1$ $a = 1$ $b = -1$ $c = 1$ $T_n = n^2 - n + 1$ $133 = n^2 - n + 1$ $n^2 - n + 1 - 133 = 0$ $n^2 - n - 132 = 0$ $(n - 12)(n + 11) = 0$ $n = 12$ or $n \neq -11$ Only the 12th message. n must be a positive whole number	✓A 14 and 26 in NEW pattern ✓A ALL values of a, b & c ✓CA general term ✓CA $T_n = 266$ ✓CA factors ✓CA both answers with rejection (6) OR ✓A 7 and 13 in NEW pattern ✓A ALL values of a, b & c ✓CA general term ✓CA $T_n = 133$ ✓CA factors ✓CA both answers with rejection (6)	

[13]

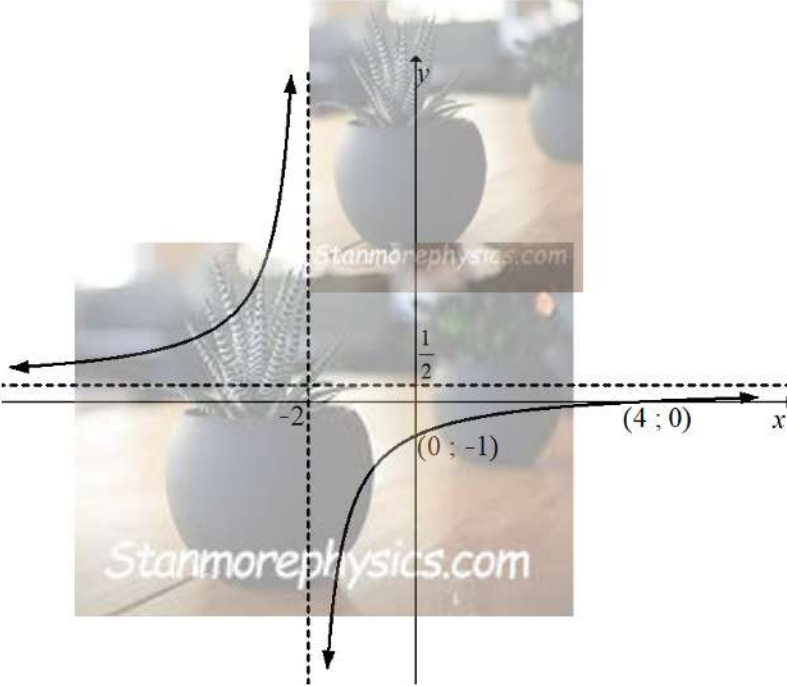
QUESTION 4

4.1.	$-7 ; m ; 5 ; n$ $(m+7) ; (5-m) ; (n-5)$first difference $-2m-2 ; m+n-10$second difference $\therefore -2m-2=4 \quad \therefore m+n-10=4$ $-2m=6 \quad -3+n-10=4$ $m=-3 \quad n=17$	4.1 is a proof, all marks are accuracy. ✓ A sec diff. in terms of m and n ✓ A $-2m=6$ ✓ A $-3+n-10=4$ (3)
4.2	$4 ; 8 ; 12$ $4n = 292$ $n = 73$ $\therefore T_{73}$ and T_{74} in a quadratic sequence	✓ A $4n = 292$ ✓ CA value of n ✓ CA T_{73} & T_{74} (3)
4.3	$T_n = 2n^2 - 2n - 7$ $1005 = 2n^2 - 2n - 7$ $2n^2 - 2n - 1012 = 0$ $n^2 - n - 506 = 0$ $(n-23)(n+22) = 0$ $n = 23$ or $n = -22$ $T_{23} = 1005$	✓ A $T_n = 1005$ ✓ A standard form ✓ CA factors ✓ CA both answers with rejection (4)
[10]		

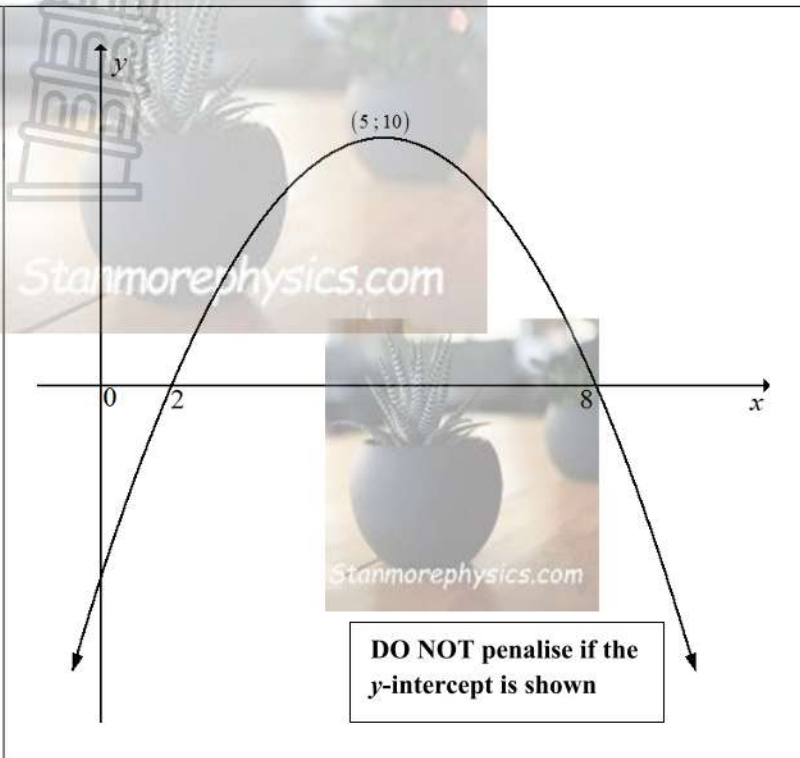
QUESTION 5

5.1	$f(x) = \frac{-3}{x+2} + \frac{1}{2}$ $f(0) = \frac{-3}{(0)+2} + \frac{1}{2}$ $y = -1$	✓ A answer (1)
5.2	$x = -2$ $y = \frac{1}{2}$	✓ A answer ✓ A answer (2)

CAPS – Grade 11
Marking Guidelines

5.3	$f(x) = \frac{-3}{x+2} + \frac{1}{2}$ $0 = \frac{-3}{x+2} + \frac{1}{2}$ $\frac{1}{2} = \frac{-3}{x+2}$ $-x-2 = -6$ $-x = -4$ $x = 4$ Answer only full marks, and ALL marks are accuracy since it is given	✓A let $y = 0$ ✓A answer (2)
5.4		✓A shape ✓CA both asymptotes ✓CA both intercepts (3)
5.5	$y > \frac{1}{2}$ or $-1 \leq y \leq 0$ OR $y \in \left(-\infty; \frac{1}{2}\right)$ or $y \in [-1; 0]$	✓✓A answer OR ✓✓A answer (2)
5.6	$p(x) = \frac{3}{x-2} + 2$ OR $p(x) = \frac{-3}{-x+2} + 2$	✓A $\frac{3}{x-2}$ ✓A +2 OR ✓A $\frac{-3}{-x+2}$ ✓A +2 (2)
[12]		

QUESTION 6

6.1		✓ A T.P (5 ; 10) ✓ A both x – intercepts. ✓ A shape (3)
6.2.1	$-x^2 + 9 = 0$ $x^2 - 9 = 0$ $(x + 3)(x - 3) = 0$ $x = -3$ or $x = 3$ $A(-3 ; 0)$ Answer only full marks OR $-x^2 + 9 = 0$ $x^2 = 9$ $x = \pm 3$ $x = -3$ or $x = 3$ $A(-3 ; 0)$ Answer only full marks	✓ A let $y = 0$ ✓ A correct factors ✓ CA answer (3) OR ✓ A let $y = 0$ ✓ A isolating the square ✓ CA answer (3)

CAPS – Grade 11
Marking Guidelines

6.2.2	Other x – intercept in h is $(9 \ 0)$ $y = a(x - x_1)(x - x_2)$ $y = a(x + 3)(x - 9)$ $-\frac{27}{4} = a(0 + 3)(0 - 9) \quad \text{OR} \quad -9 = a(3 + 3)(3 - 9)$ $-\frac{27}{4} = a(-27) \quad \text{OR} \quad -9 = -36a$ $a = \frac{1}{4}$ $\therefore y = \frac{1}{4}(x + 3)(x - 9)$ $\therefore y = \frac{1}{4}(x^2 - 6x - 27)$ $\therefore y = \frac{1}{4}x^2 - \frac{3}{2}x - \frac{27}{4}$ $a = \frac{1}{4} \quad \text{and} \quad b = -\frac{3}{2}$ OR $C(3; -9)$ $y = a(x + p)^2 + q$ $y = a(x - 3)^2 - 9$ $0 = a(-3 - 3)^2 - 9 \quad \text{OR} \quad -\frac{27}{4} = a(0 - 3)^2 - 9$ $9 = a(36) \quad \text{OR} \quad \frac{9}{4} = 9a$ $a = \frac{1}{4}$ $y = \frac{1}{4}(x - 3)^2 - 9$ $y = \frac{1}{4}(x^2 - 6x + 9) - 9$ $y = \frac{1}{4}x^2 - \frac{3}{2}x - \frac{27}{4}$ $\therefore a = \frac{1}{4} \quad \text{and} \quad b = -\frac{3}{2}$ Since 6.2.2 is a proof, all marks are accuracy.	✓ A identifying $(9; 0)$ ✓ A substitution of x – intercepts ✓ A subst. of $\left(0; -\frac{27}{4}\right)$ or $(3; -9)$ ✓ A simplification of h. OR ✓ A identifying $C(3; -9)$ ✓ A substitution of $C(3; -9)$ ✓ A subst. $\left(0; -\frac{27}{4}\right)$ or $(-3; 0)$ ✓ A simplification to h. (4)
6.2.3	$C(3; -9)$ $y \in \mathbb{R} : y \geq -9$ OR $C(3; -9)$ $y \in [-9; \infty)$ Answer only full marks Answer only full marks	✓ A $y_c = -9$ ✓ A answer OR ✓ A $y_c = -9$ ✓ A answer (2)

CAPS – Grade 11
Marking Guidelines

6.2.4	Reflection about x-axis, and shift of 5 units up OR shift of 5 units down, and reflection about the x-axis	Must be in this order Must be in this order	✓ A reflection about the x-axis ✓ A 5 units up (2) OR ✓ A 5 units down ✓ A reflection about the x-axis (2)
6.2.5	$\frac{1}{4}x^2 - \frac{3}{2}x - \frac{27}{4} = -x^2 + 9$ $\frac{5}{4}x^2 - \frac{3}{2}x - \frac{63}{4} = 0$ $5x^2 - 6x - 63 = 0$ $(x+3)(5x-21) = 0$ $x = -3 \text{ or } x = \frac{21}{5} = 4,2$ $y = 0 \text{ or } y = -\frac{216}{25} = -8,64$ $D(4,2 ; -8,64)$		✓ A equating both equations ✓ CA standard form ✓ CA correct factors ✓ CA value of x ✓ CA value of y (5)
6.2.6	$h(-1) = -5$ & $h(7) = -5$ $\therefore k = -5$	Answer only full marks	✓ A x-value of -1 or 7 ✓ A answer (2)
[21]			

QUESTION 7

7.1	$h(x) = 2^x + q$ $-7\frac{3}{4} = 2^{-2} + q$ $q = -8$		✓ A substituting $\left(-2 ; -7\frac{3}{4}\right)$ ✓ A answer (2)
7.2	$2^x - 8 = 0$ $2^x = 2^3$ $x = 3$ A(3 ; 0)	Answer only full marks	✓ A let $y = 0$ ✓ A $2^x = 2^3$ ✓ A answer (3)

CAPS – Grade 11
Marking Guidelines

7.3	$f(0) = 2^0 - 8$ $f(0) = -7$ $B(0; -7)$ $m = \frac{y_2 - y_1}{x_2 - x_1}$ $m = \frac{0 - (-7)}{3 - 0}$ $m = \frac{7}{3}$	✓ A coordinate of B ✓ CA substitution ✓ CA answer (3)
7.4	$0 \leq x < 3$ OR $x \in [0; 3)$	✓✓ AA answer (2) OR ✓✓ AA answer (2)
7.5	$7 < k < 8$ OR $k \in (7; 8)$	✓✓ CACA answer (2) OR ✓✓ CACA answer (2)
[12]		

QUESTION 8

8.1.	$A = P(1 + i \cdot n)$ $A = R 15\,000(1 + 0,10 \times 6)$ $A = R 24\,000$ Interest = R24 000 – R15 000 = R 9000 OR Interest = $p \times i \times n$ $= 15\,000 \times 0,1 \times 6$ $= R 9000$	✓ A subst. into correct formula ✓ A value of A ✓ CA answer (3) OR ✓ A correct formula ✓ A subst. into correct formula ✓ CA answer (3)
8.2	$A = P(1 - i)^n$ $R 277\,216 = P(1 - 0,0825)^3$ $\frac{455625}{(1 - 0,0825)^3} = P$ $\therefore P = R 358\,922$	✓ A subst. to correct formula ✓ CA answer (2)

CAPS – Grade 11
Marking Guidelines

8.3	$1 + i_{eff} = \left(1 + \frac{i_{nom}}{m}\right)^m$ $1 + 0,12 = \left(1 + \frac{i_{nom}}{4}\right)^4$ $i_{nom} = 4 \left(\sqrt[4]{1,12} - 1\right) \times 100\%$ $r = 11,49\%$	✓ A correct formula ✓ A subst into correct formula ✓ CA simplification ✓ CA answer (4)
8.4	$A = P(1+i)^n$ $A = R25\,0000 \left(1 + \frac{0,0925}{2}\right)^{19}$ $A = R59\,022$	✓ A $\frac{0,0925}{2}$ ✓ $n = 19$ ✓ CA answer (3)
8.5	$6500 \left(1 + \frac{0,13}{12}\right)^{12 \times 2} \left(1 + \frac{0,105}{4}\right)^{4 \times 3} - x \left(1 + \frac{0,105}{4}\right)^{4 \times 2} = 7\,428,30$ $x = \frac{6500 \left(1 + \frac{0,13}{12}\right)^{24} \left(1 + \frac{0,105}{4}\right)^{12} - 7\,428,30}{\left(1 + \frac{0,105}{4}\right)^8}$ $x = R3\,300$ OR $\left[6500 \left(1 + \frac{0,13}{12}\right)^{24} \cdot \left(1 + \frac{0,105}{4}\right)^4 - x\right] \left(1 + \frac{0,105}{4}\right)^8 = 7\,428,30$ $-x = \frac{7\,428,30}{\left(1 + \frac{0,105}{4}\right)^8} - 6500 \left(1 + \frac{0,13}{12}\right)^{24} \cdot \left(1 + \frac{0,105}{4}\right)^4$ $-x = -R\,3300$ $\therefore x = R3\,300$ OR	✓ A $6500 \left(1 + \frac{0,13}{12}\right)^{24}$ ✓ A $\left(1 + \frac{0,105}{4}\right)^{12}$ ✓ A $-x \left(1 + \frac{0,105}{4}\right)^8$ ✓ A all equated to 7 428,30 ✓ CA answer (5) OR ✓ A $6500 \left(1 + \frac{0,13}{12}\right)^{24}$ ✓ A $\left(1 + \frac{0,105}{4}\right)^4 - x$ ✓ A $\left(1 + \frac{0,105}{4}\right)^8$ ✓ A ALL equated to 7 428,30 ✓ CA answer (5) OR

CAPS – Grade 11
Marking Guidelines

$$A_3 = 6500 \left(1 + \frac{0,13}{12}\right)^{24} \cdot \left(1 + \frac{0,105}{4}\right)^4 - x \dots \text{forward equation}$$

$$P_3 = \frac{7428,30}{\left(1 + \frac{0,105}{4}\right)^8} \dots \text{reverse equation}$$

$$A_3 = P_3$$

$$6500 \left(1 + \frac{0,13}{12}\right)^{24} \cdot \left(1 + \frac{0,105}{4}\right)^4 - x = \frac{7428,30}{\left(1 + \frac{0,105}{4}\right)^8}$$

$$-x = \frac{7428,30}{\left(1 + \frac{0,105}{4}\right)^8} - 6500 \left(1 + \frac{0,13}{12}\right)^{24} \cdot \left(1 + \frac{0,105}{4}\right)^4$$

$$-x = -R\ 3300$$

$$\therefore x = R3300$$

OR

$$A_3 = 6500 \left(1 + \frac{0,13}{12}\right)^{24} \cdot \left(1 + \frac{0,105}{4}\right)^4 \dots \text{forward equation}$$

$$P_3 = \frac{7428,30}{\left(1 + \frac{0,105}{4}\right)^8} + x \dots \text{reverse equation}$$

$$A_3 = P_3$$

$$6500 \left(1 + \frac{0,13}{12}\right)^{24} \cdot \left(1 + \frac{0,105}{4}\right)^4 = \frac{7428,30}{\left(1 + \frac{0,105}{4}\right)^8} + x$$

$$x = 6500 \left(1 + \frac{0,13}{12}\right)^{24} \cdot \left(1 + \frac{0,105}{4}\right)^4 - \frac{7428,30}{\left(1 + \frac{0,105}{4}\right)^8}$$

$$\therefore x = R3300$$

$$\checkmark A \quad 6500 \left(1 + \frac{0,13}{12}\right)^{24}$$

$$\checkmark A \quad \left(1 + \frac{0,105}{4}\right)^4 - x$$

$$\checkmark A \quad \left(1 + \frac{0,105}{4}\right)^8$$

$$\checkmark A \quad \text{ALL equated to } 7428,30$$

$\checkmark$ CA answer

(5)

OR

$$\checkmark A \quad 6500 \left(1 + \frac{0,13}{12}\right)^{24}$$

$$\checkmark A \quad \left(1 + \frac{0,105}{4}\right)^4 - x$$

$$\checkmark A \quad \left(1 + \frac{0,105}{4}\right)^8$$

$$\checkmark A \quad \text{ALL equated to } 7428,30$$

$\checkmark$ CA answer

(5)

[17]

QUESTION 9

9.1.1	$P(A \text{ and } C) = 0$	✓ A answer (1)
9.1.2	$P(A \text{ or } C) = P(A) + P(C) - P(A \text{ and } C)$ $P(A \text{ or } C) = 0,47 + \frac{3}{7} - 0$ $P(A \text{ or } C) = 0,90$ OR $P(A \text{ or } C) = P(A) + P(C)$ $P(A \text{ or } C) = 0,47 + \frac{3}{7}$ $P(A \text{ or } C) = 0,90$	✓ A substitution to a correct rule ✓ A answer OR ✓ A substitution to a correct rule ✓ A answer (2)
9.2.1	$P(A \text{ and } B) = P(A) \times P(B)$ $P(A \text{ and } B) = \frac{2}{5} P(B) \times P(B)$ $P(A \text{ and } B) = \frac{2}{5} P(B)^2$	✓ A $P(A \text{ and } B) = P(A) \times P(B)$ ✓ A answer (2)
9.2.2	$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$ $0,7 = \frac{2}{5} P(B) + P(B) - \frac{2}{5} P(B)^2$ $0,7 = \frac{7}{5} P(B) - \frac{2}{5} P(B)^2$ $\frac{2}{5} P(B)^2 - \frac{7}{5} P(B) + \frac{7}{10} = 0$ OR $4P(B)^2 - 14P(B) + 7 = 0$ $P(B) = \frac{-\left(-\frac{7}{5}\right) \pm \sqrt{\left(-\frac{7}{5}\right)^2 - 4\left(\frac{2}{5}\right)\left(\frac{7}{10}\right)}}{2\left(\frac{2}{5}\right)}$ $P(B) = \frac{-(-14) \pm \sqrt{(-14)^2 - 4(4)(7)}}{2(4)}$ $P(B) = 0,60 \quad \text{or} \quad P(B) \neq 2,90$ OR	✓ CA substitution to correct rule ✓ CA equation in terms $P(B)$ ✓ CA substitution into quadratic formula ✓ CA answer OR < (5)

If answer is CA,
 $0 < P(B) < 1$

CAPS – Grade 11
Marking Guidelines

$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$ $0,7 = P(A) + \frac{5}{2}P(A) - P(A) \cdot \frac{5}{2}P(A)$ $0,7 = \frac{7}{2}P(A) - \frac{5}{2}P(A)^2$ $\frac{5}{2}P(A)^2 - \frac{7}{2}P(A) + \frac{7}{10} = 0 \quad \text{OR} \quad 25P(A)^2 - 35P(A) + 7 = 0$ $P(A) = \frac{-\left(-\frac{7}{2}\right) \pm \sqrt{\left(-\frac{7}{2}\right)^2 - 4\left(\frac{5}{2}\right)\left(\frac{7}{10}\right)}}{2\left(\frac{5}{2}\right)}$ $P(A) = \frac{-(-35) \pm \sqrt{(-35)^2 - 4(25)(7)}}{2(25)}$ $P(A) = 0,24 \quad \text{or} \quad P(A) \neq 1,16$ $P(A) = \frac{2}{5}P(B)$ $0,24 = \frac{2}{5}P(B)$ $\therefore P(B) = 0,60 \quad \text{or} \quad \frac{3}{5}$ If answer is CA, $0 < P(B) < 1$	✓CA substitution to correct rule ✓CA equation in terms of $P(A)$ ✓CA substitution into quadratic formula ✓CA $P(A)$ ✓CA answer (5)
---	---

[10]

QUESTION 10

10.1.1

Let the number of females who will love to buy Gucci be x

	Gucci	Louis Vuitton	
Males	y	$2x$	45
Females	x	$3y$	85
	$x + y$	$2x + 3y$	130

$$y + 2x = 45 \dots \text{line 1}$$

$$x + 3y = 85 \dots \text{line 2}$$

$$y = 45 - 2x \dots \text{OR} \quad x = 85 - 3y$$

$$x + 3(45 - 2x) = 85 \quad \text{OR} \quad y + 2(85 - 3y) = 45$$

$$x + 135 - 6x = 85 \quad \text{OR} \quad y + 170 - 6y = 45$$

$$-5x = -50 \quad \text{OR} \quad -5y = -125$$

$$x = 10 \quad y = 25$$

$$x = 10$$

x must be a positive whole number

$$\checkmark \text{A males : } y + 2x = 45$$

$$\checkmark \text{A females : } x + 3y = 85$$

$$\checkmark \text{A } y = 45 - 2x \quad \text{OR} \quad x = 85 - 3y$$

$\checkmark$ CA substitution

$\checkmark$ CA simplification

$\checkmark$ CA answer

(6)

CAPS – Grade 11
Marking Guidelines

OR

	Gucci	Louis Vuitton	
Males	y	$45 - y$	45
Females	$85 - 3y$	$3y$	85
	$85 - 2y$	$45 + 2y$	130

$$45 - y = 2(85 - 3y)$$

$$45 - y = 170 - 6y$$

$$5y = 125$$

$$y = 25$$

$x = 10$ females who would love Gucci items



OR

Allow n to represent total number

$$n(M) = 85 \quad \& \quad n(F) = 45$$

$$n(M \text{ and } G) = y$$

$$n(M \text{ and } LV) = 2(85 - 3y) = 170 - 6y$$

$$n(F \text{ and } G) = 85 - 3y$$

$$n(F \text{ and } LV) = 3y$$

$$\therefore 2(85 - 3y) + y = 45$$

$$170 - 6y + y = 45$$

$$-5y = -125$$

$$y = 25$$

$x = 10$ females who would love Gucci items

OR

$$\checkmark A \quad M \text{ and } LV = 45 - y$$

$$\checkmark A \quad F \text{ and } Gucci = 85 - 3y$$

$$\checkmark A \quad 45 - y = 2(85 - 3y)$$

$$\checkmark CA \quad \text{values of } y$$

$$\checkmark CA \quad \text{answer}$$

(6)

OR

$$\checkmark A \quad 2(85 - 3y) \text{ or } 170 - 6y$$

$$\checkmark A \quad 85 - 3y$$

$$\checkmark A \quad 2(85 - 3y) + y = 45$$

$$\checkmark CA \quad \text{value of } y$$

$$\checkmark CA \quad \text{answer.}$$

(6)

10.2

$$P(M \text{ and } G) = \frac{25}{130}$$

$$= 0,19$$

$$P(M) \times P(G) = \frac{45}{130} \times \frac{35}{130}$$

$$= \frac{63}{675}$$

$$= 0,09$$

$$P(M \text{ and } G) \neq P(M) \times P(G)$$

Both events are NOT independent

$$\checkmark CA \quad \frac{25}{130} \text{ or } \frac{5}{26} \text{ or } 0,19$$

$$\checkmark CA \quad \frac{45}{130} \times \frac{35}{130}$$

$$\checkmark CA \quad \frac{63}{675} \text{ or } \frac{7}{75} \text{ or } 0,09$$

$$\checkmark CA \quad \text{NOT independent}$$

(4)

CAPS – Grade 11
Marking Guidelines

OR $P(M \text{ and } LV) = \frac{20}{130}$ $= 0,15$ $P(M) \times P(LV) = \frac{45}{130} \times \frac{95}{130}$ $= \frac{171}{676}$ $= 0,25$ $P(M \text{ and } LV) \neq P(M) \times P(LV)$ Both events are NOT independent	OR $\checkmark \text{CA } \frac{20}{130} \text{ or } \frac{2}{13} \text{ or } 0,19$ $\checkmark \text{CA } \frac{45}{130} \times \frac{95}{130}$ $\checkmark \text{CA } \frac{171}{676} \text{ or } 0,25$ $\checkmark \text{CA NOT independent}$ (4)
[10]	

TOTAL: 150